Capacity and departure scheduling in MATSim

LSGI3213 Intelligent Transportation Systems

A reproducible experiment with the official equil network

Updated results: sensitivity, convergence and score mechanisms

Research questions

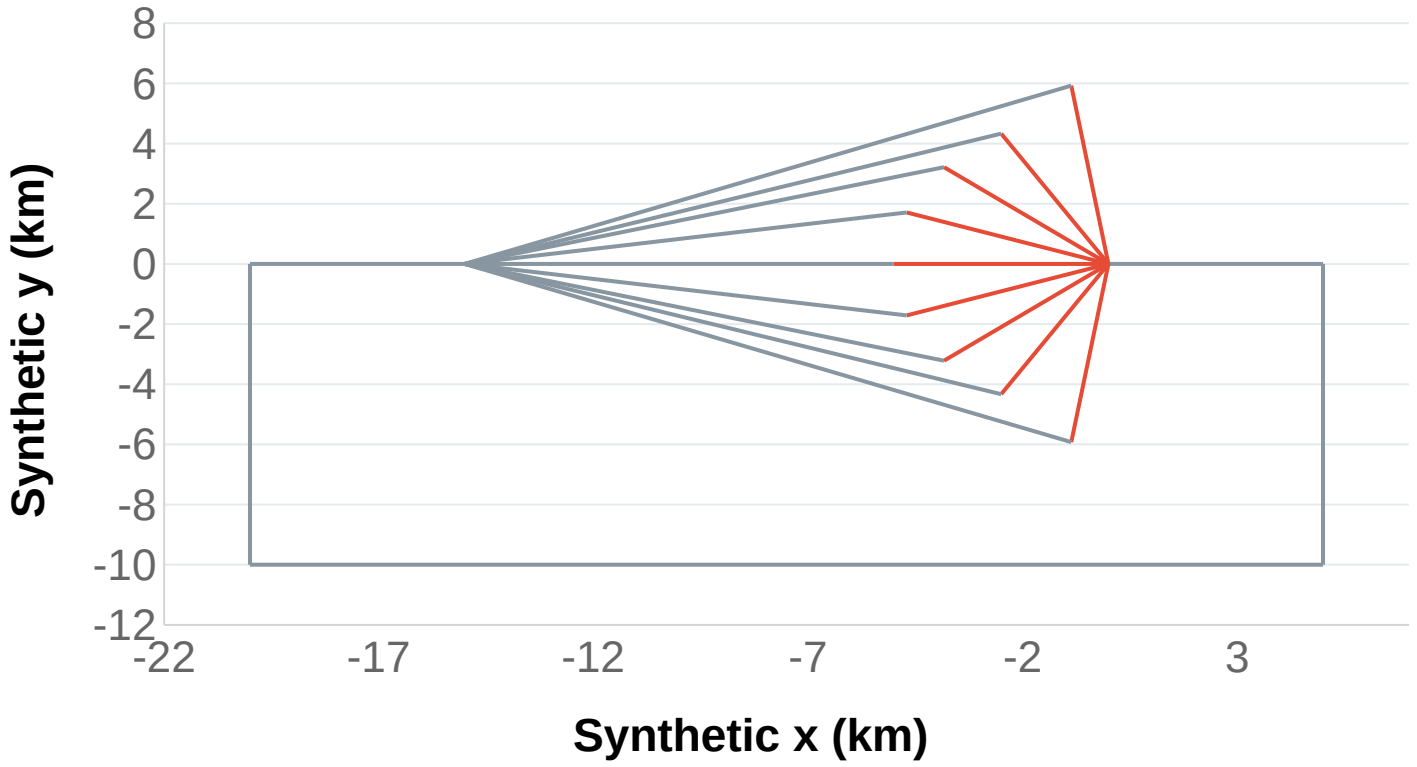
What changes when bottleneck capacity doubles?

How much departure spreading removes congestion?

Do capacity and scheduling provide overlapping benefits?

Why can faster journeys still produce lower daily scores?

Official equil data



15 nodes; 23 directed links

2,000 agents: home-work-home

Nine outward corridors; shared return path

Downloaded inputs match the tutorial files byte for byte

Source: MATSim matsim-12.0 equil example; public repository and SHA-256 manifest included

MATSim execution and learning

Select a remembered daily plan

Execute all agents together in QSim

Score activities, travel and timing

ReRoute explores alternatives; BestScore selects remembered plans

Sources: MATSim Book; MATSim 12.0 configuration and Java source

Five controlled scenarios

Scenario	Supply / schedule	Replanning
Baseline	All first departures at 06:00	BestScore 0.9; ReRoute 0.1
Capacity	Links 11-19: 1,000 → 2,000 veh/h	Same
Stagger	Departures spread 05:30-06:30	Same
Combined	Capacity + Stagger	Same
Adaptive	Baseline initial network and plans	0.1 time mutation; BestScore 0.8

Common controls: scoring, demand, facilities, free speed, lanes, iterations and seed set

Execution and independent checks

15 original main runs plus 18 new simulation runs

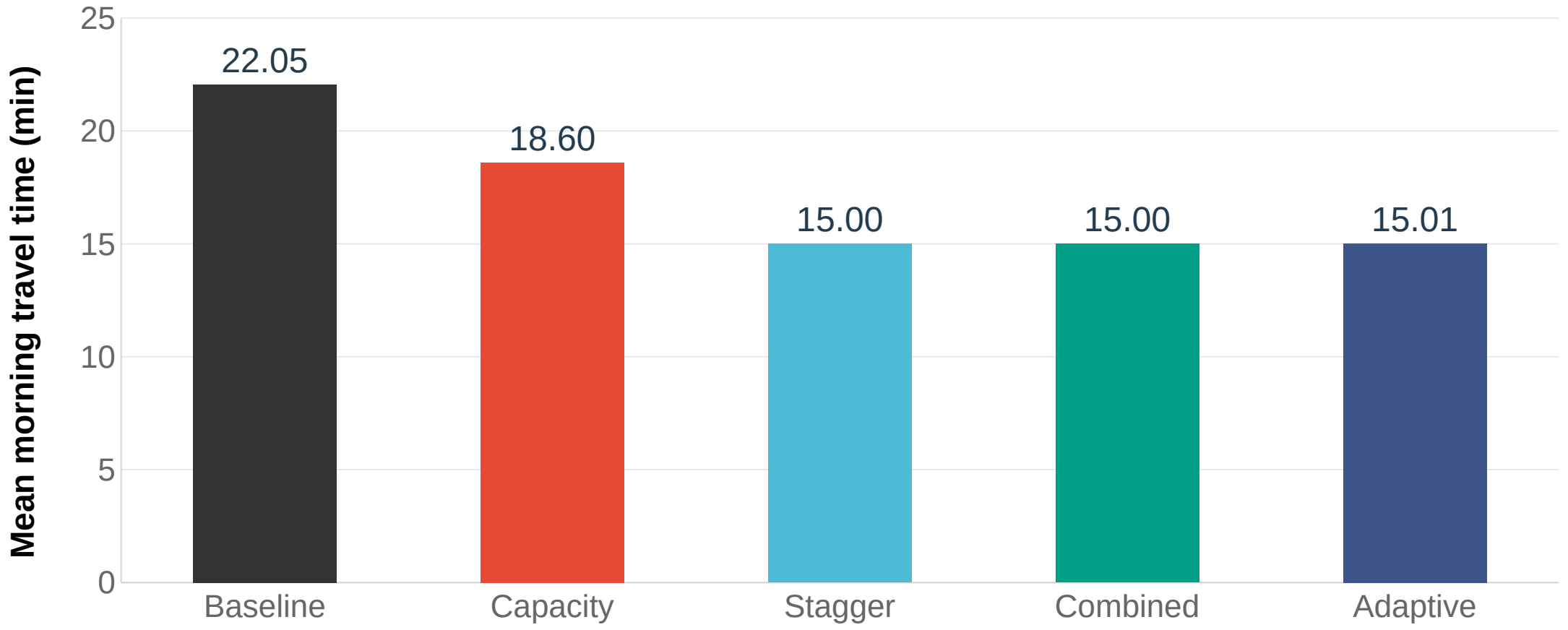
Capacity/window grid: 27 records, including 12 reused runs

Adaptive extension: three seeds, iterations 0-500

All 30 supplementary records: 4,000 trips and zero stuck agents

MATSim 12.0; original Java 11, new runs Java 14; events, trips and scores cross-checked

Morning travel time



Iteration 100 reference. Mean of three final-iteration runs; 2,000 morning trips per run; bar axis starts at zero

Capacity benefit depends on departure spreading

Morning travel time in minutes, averaged over three seeds

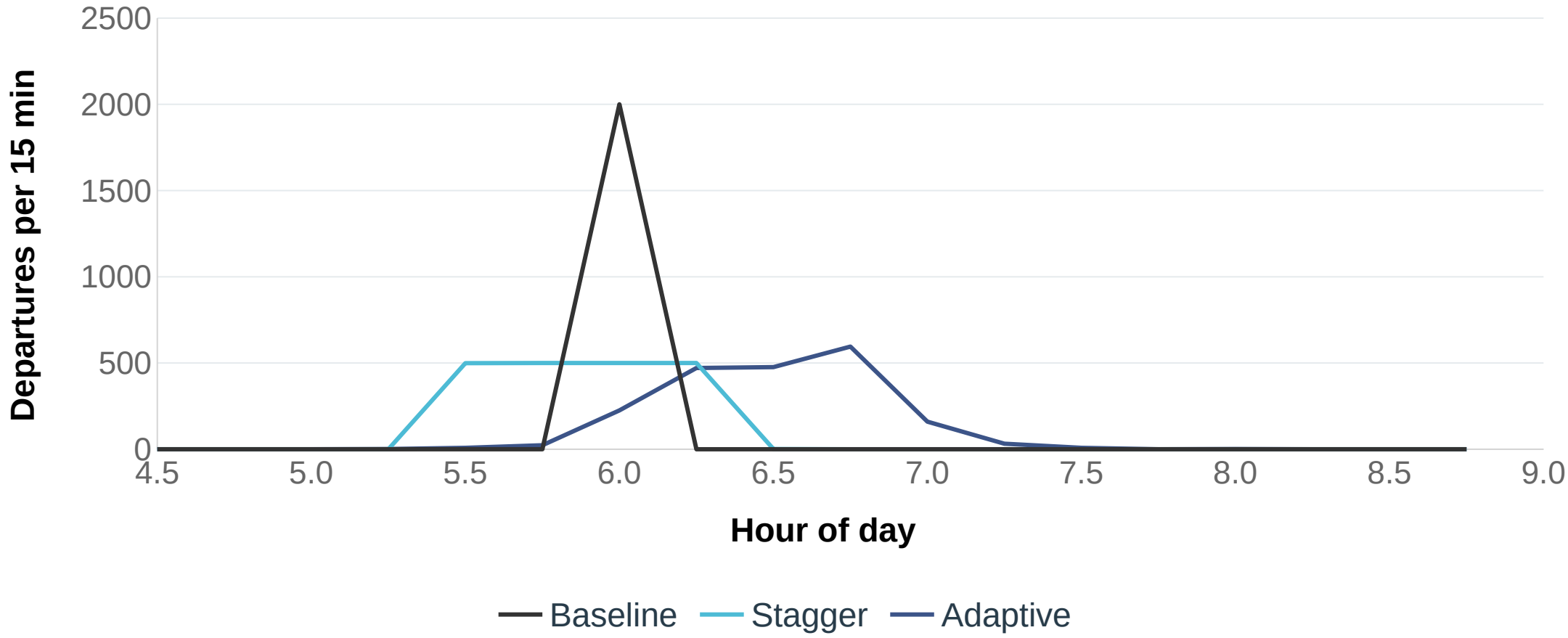
Corridor capacity	0 min window	30 min window	60 min window
1x	22.052	15.019	15.002
1.5x	19.729	15.002	15.000
2x	18.599	15.001	15.000

30 to 60 minutes of spreading saves only another 1.06 seconds

Doubling capacity saves 207 seconds with simultaneous departures,
but only 1.11 seconds with a 30-minute window

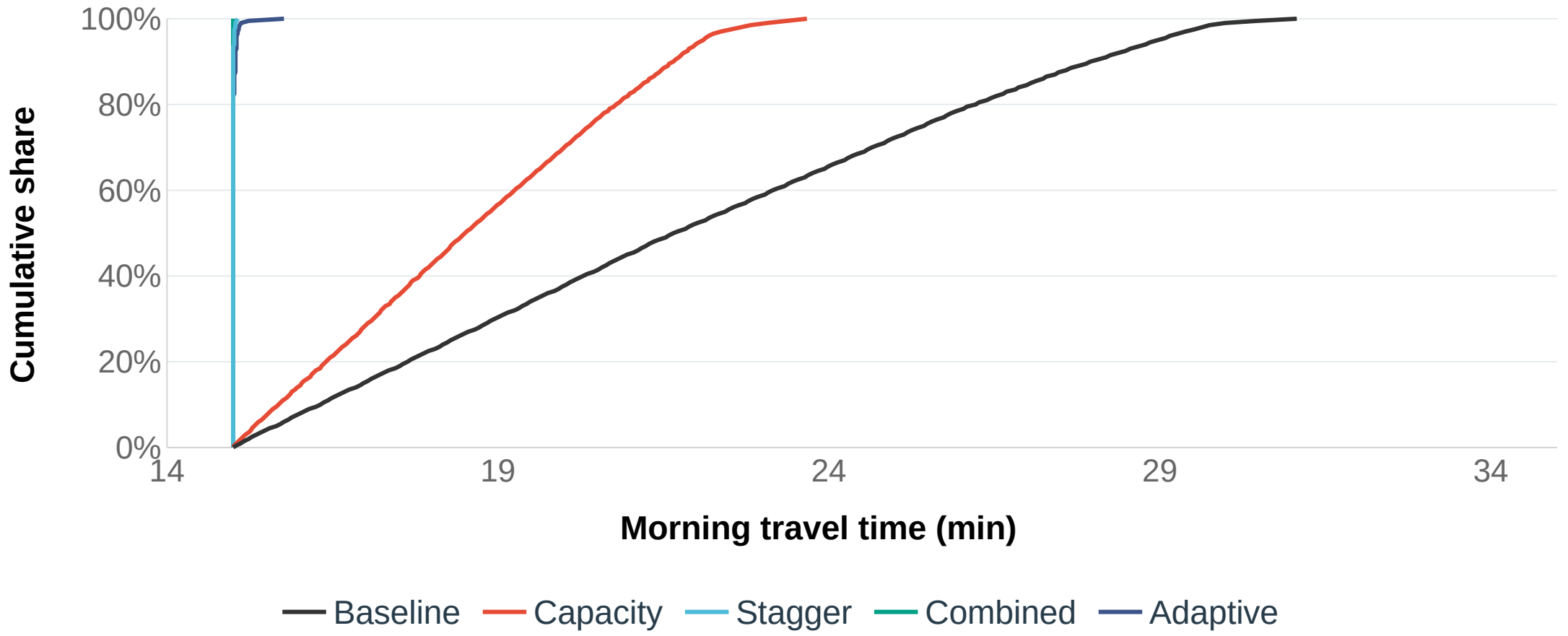
3 capacities × 3 windows × 3 seeds. Synthetic demand, 2,000 agents. Free-flow travel is about 15 min.

Realised departure patterns



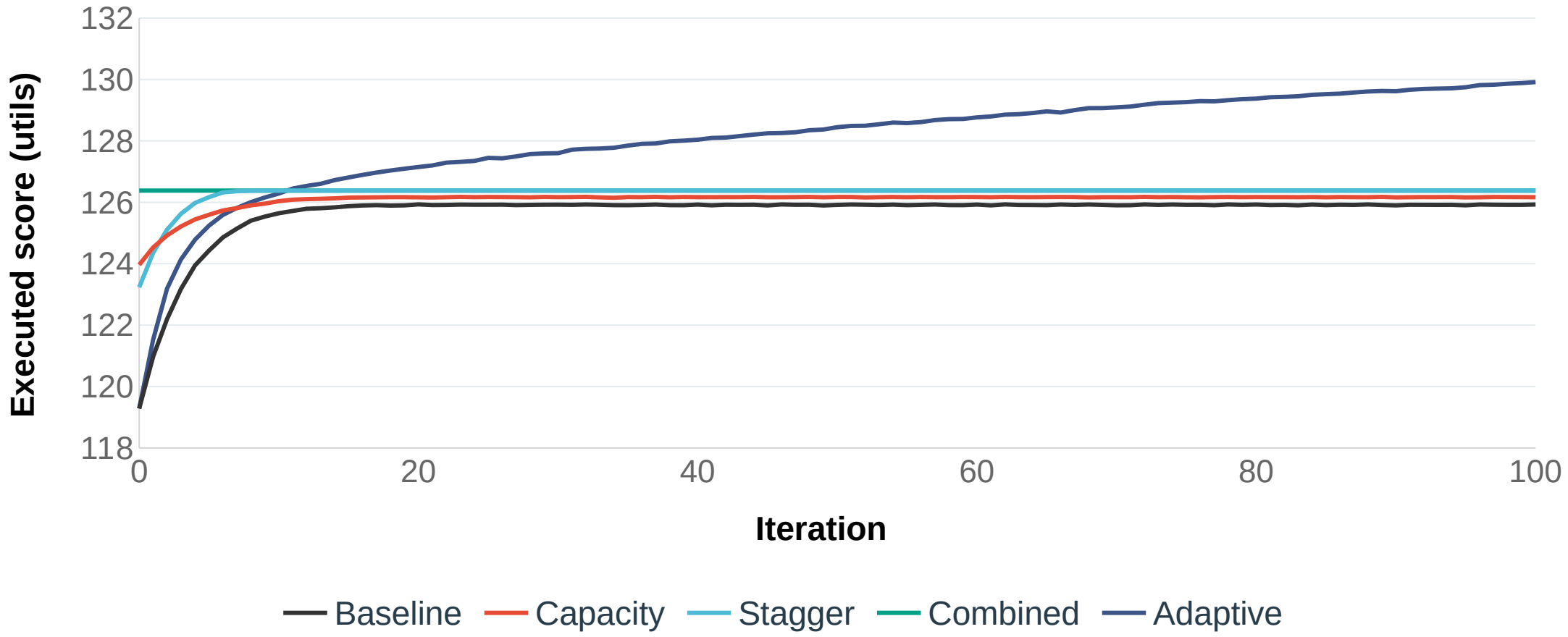
Iteration 100 reference. Seed 4711; matching 5-minute histogram is supplied in the report and figure data

Travel-time distributions



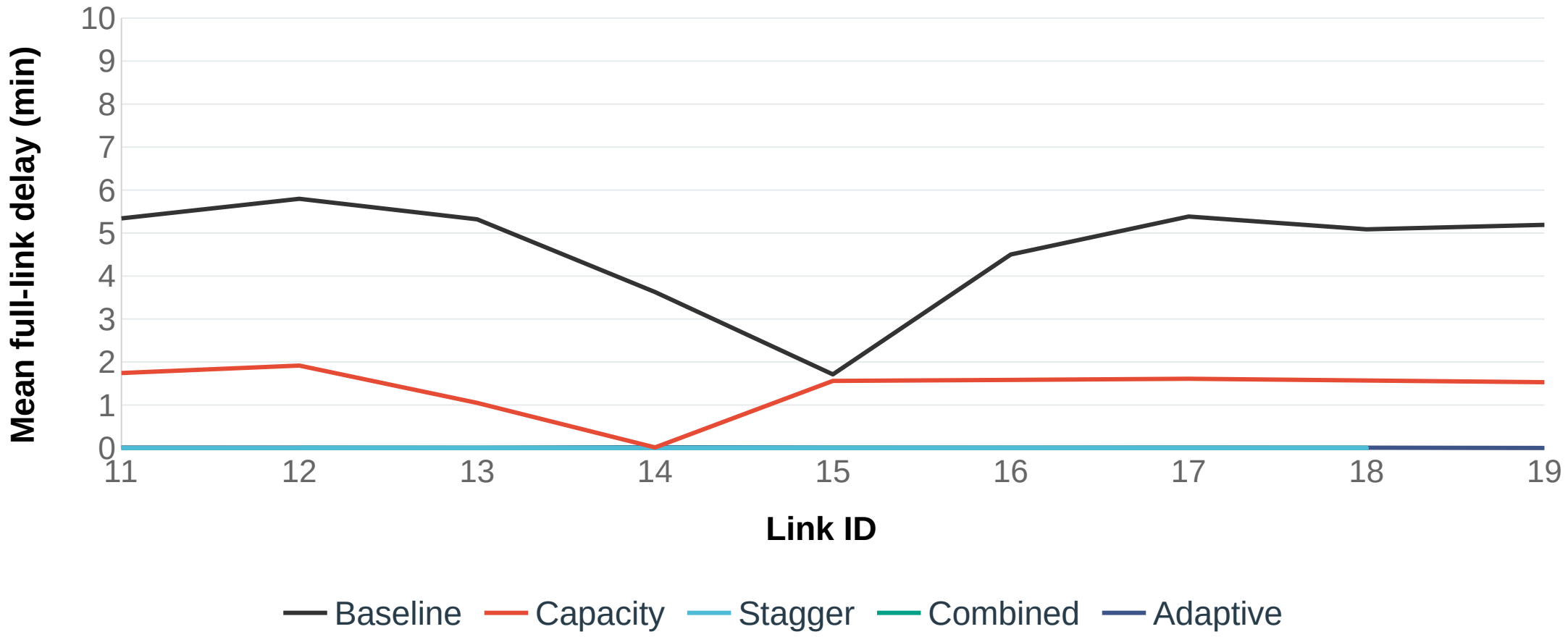
Iteration 100 reference. Seed 4711; 2,000 morning trips per scenario; ordered points sampled for display

Executed scores and timing preferences



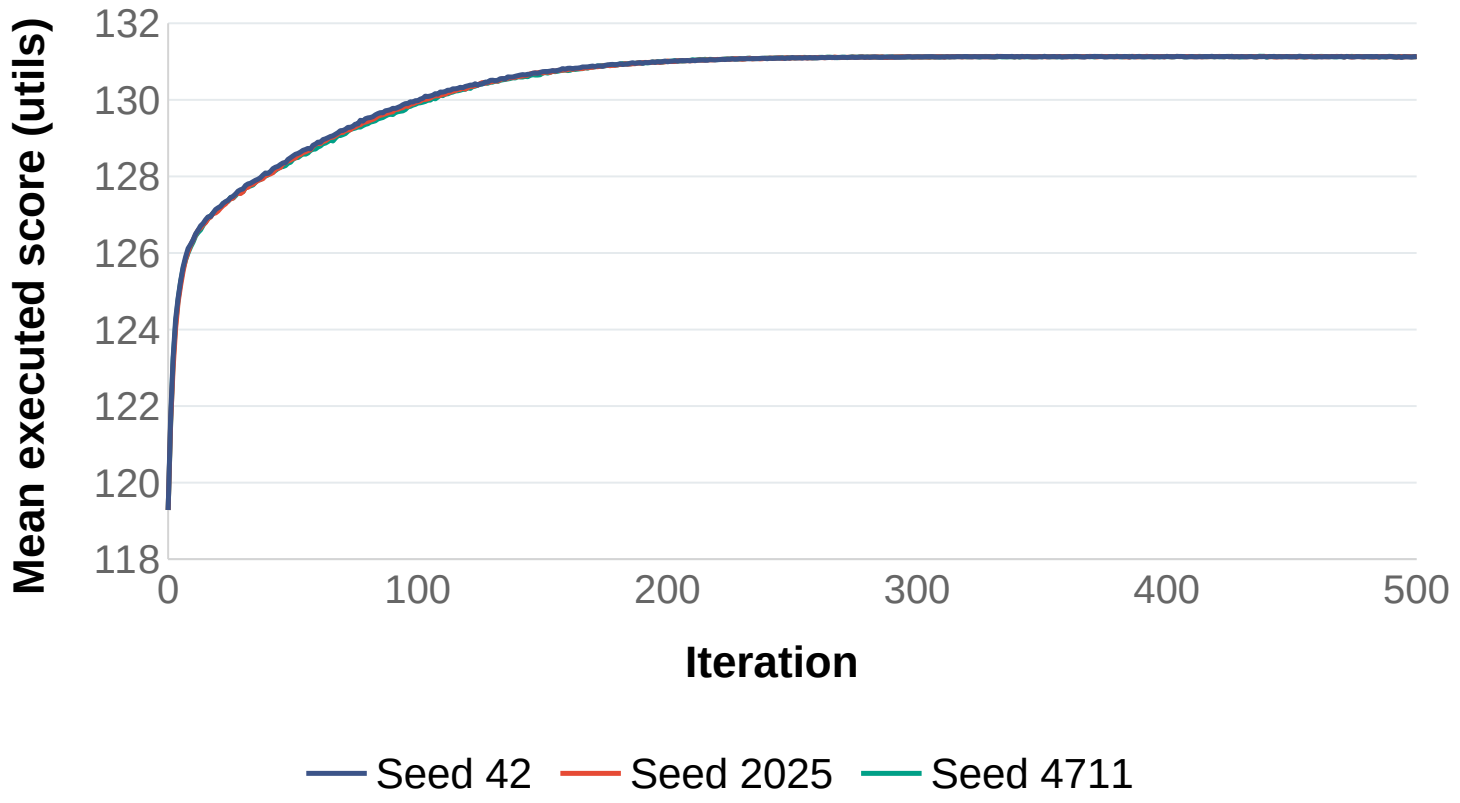
Iteration 100 reference. Seed 4711 shown; identical scoring rules; 500-iteration extension follows

Delay on the nine corridor links



Iteration 100 reference. Seed 4711; delay = max(traversal time − length/free speed, 0); endpoint links excluded

Adaptive scores stabilise by iteration 500



All 3 seeds pass the predefined stability screen

Final mean changes: 0.0011-0.0020 utils

Threshold: 0.05 utils
Descriptive stability check, not proof of equilibrium

Compare iterations 451-500 against 401-450. At iteration 100, adaptive learning still continued.

Work-activity score losses outweigh travel gains

Mean score changes among people whose daily score falls (utils)

Window	Lower score	Travel	Home	Work	Net
30 min	28.22%	+1.176	+0.941	-2.465	-0.349
60 min	41.43%	+1.112	+0.889	-3.193	-1.192

Every score loser gains travel and home-activity utility

Work scoring excludes time before 07:00. The direct waiting penalty is zero.

9 runs, 18,000 person-run records. Reconstructed totals match native scores within 6×10^{-14} utils.

Earlier arrival also brings an earlier work departure

Person 1367, seed 4711	Baseline	60 min window
Arrival at work	06:26:09	05:57:18
Departure from work	14:26:09	13:57:18
Presence at work	8 hours	8 hours
Work time used in scoring	7:26:09	6:57:18

The plan ends work 8 hours after arrival, while scoring starts at 07:00

Actual simulated record. Work utility falls 3.209 utils, exceeding travel (+1.115) and home (+0.892) gains.

Interpretation for ITS

All tested windows retain 2,000 agents and the same scoring rules

30 minutes nearly removes queues under the current demand

Lower scores reflect the work schedule and opening-time interaction

Real applications require observed demand and feasible work schedules

Conclusions

A 30-minute window nearly reaches the 15-minute free-flow time

Extra capacity adds little after departure spreading removes queues

Adaptive learning needs more than the original 100 iterations

Travel gains can coexist with losses in daily activity utility

Sources and acknowledgment

MATSim developers: official equil inputs, matsim-12.0 tag

Horni, Nagel & Axhausen (2016), The Multi-Agent Transport Simulation MATSim

OpenAI Codex assisted with data discovery, code, execution, analysis, report and slides

Updated data, scoring-source extracts and reproducible analysis accompany this version

github.com/matsim-org/matsim-libs | doi.org/10.5334/baw